Applied Agentic AI: Metacognition, Verification, and "In-the-Loop" Learning

Proposed course syllabus · graduate and undergraduate versions

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Course description

Metacognition is thinking about thinking. An AI agent adds a second layer. We have to think about how the agent thinks, so that it scaffolds our work and does not replace our judgment.

AI agents now draft, calculate, write code and build real work. They are fast and fluent, but they never hold the full context of a real problem, and a fluent wrong answer looks just like a right one. A person has to decide what the agent needs to know, check what it returns, and stay in the loop.

This course teaches that job. Students learn to direct an AI agent on a real problem in their own field and to stand behind the result. The methods are sustainable: each correction becomes a standing rule, regulation or exemplar, so the work keeps improving as the models change. Scaffolding means the standards, checks and approvals that surround the agent's work.

The course comes from my own experience directing AI agents in research and teaching. It is open to engineering, business and analytics students.

Who it is for

- **Graduate version:** master's and early doctoral students. Prerequisite: one course in quantitative methods, statistics or analytics.
- **Undergraduate version:** juniors and seniors. Prerequisite: one course in statistics or quantitative methods.
- No programming course or prior AI experience is required. Students learn to read and test the short scripts the agent drafts.
- Students use any AI assistant or coding agent that can read a project's files, follow standing instructions and run short scripts, such as Claude or Claude Code, ChatGPT or Codex, Google Gemini, or GitHub Copilot. No outcome depends on one product.

Learning outcomes

Students who complete the course will be able to:

1. Explain how an AI agent reaches its output, name the context it lacks, and supply that context in writing.
2. Verify a result before using or signing it, with checks that each end in a verdict: accept, reject or escalate.
3. Build the standards an agent must meet (rules, regulations and exemplars) and improve them through a learning loop.
4. Plan and govern agentic work so that a person stays in the loop for every important decision.

Modules

The six modules build on each other. Each one adds a layer to the term project, so by the end the project carries everything the course teaches.

1. Thinking about how the AI thinks. What the agent reads, what it never sees, and why material buried in a long session gets lost. Students correct a deliberately flawed analysis. *Adds to the project:* a correction log, which every later module draws on.

2. Verifying results. The nine checks, in three families. Screens run on every result: units, bounds and assumptions. Expectations run on what is used: a prior estimate, a nudge or a known landmark. Confirmations run on what is signed: triangulate, balance or source. A negative result needs a positive control. *Adds:* a verdict on every result the agent returns.

3. Rules, regulations and exemplars. Rules are checks, written as scripts, that the work must pass before it is accepted. A rule is trusted only after it has been made to fail. Regulations are written guidance for matters of judgment, given to the agent at the start of every session. Exemplars are earlier work to follow. *Builds on 1 and 2:* the learning loop turns each logged correction and each failed check into a rule, a regulation, an exemplar or a one-off

exception. *Adds:* the project's standards file.

4. Sources. Keep the source itself on file, check that the extracted text is real text, and have a working link before writing any citation. An abstract or a summary is not the paper. *Builds on 3:* each source check becomes a rule. *Adds:* a source audit.

5. Planning before building. Write a charge, take stock of what already exists, and ask clarifying questions one at a time. Then write a plan, have a critic attack it, and get a person's approval before anything is built. Build, then verify the result against the plan. *Builds on 1 to 4:* the plan names the context, checks, rules and sources the work needs. *Adds:* the approved plan.

6. The Council of Agents, and checks and balances. Several agents, each with a distinct role, analyze one question on their own. A chair reports where they agree and where they split. Every seat runs on the same model, so agreement shows breadth of framing, not proof. *Builds on all five:* the council reviews the plan, and the rules from Module 3 become gates where a person must approve. *Adds:* a map of the gates, who may override each one, and the record of what was checked.

Term project

Each student finds a real problem in an organization, such as a business, a nonprofit, a team at work or a campus office. Students weigh two or three candidates, pick the one with the biggest payoff, and work it with an agent through all six modules. Undergraduates may take an instructor-supplied problem with known flaws instead. A graduate student who has no client by the end of Module 1 takes an instructor-supplied problem and also builds a flawed artifact, with an answer key, for a classmate.

The project is built on its rules. Each time the agent gets something wrong, the student records the fix in the standards file as a rule, a regulation, an exemplar or a one-off exception. By the end, the agent's work must pass those rules, and each rule has been shown to fail on a planted case.

The project ends in a three-part defense. Students find the defects planted in a copy of their own work, on paper and without AI. They run one of their own rules and show it refusing a planted case. Then they answer one question on why a gate sits where it does.

How the versions differ

	Graduate	Undergraduate
Project	Real client required (fallback above)	Real client or instructor-supplied problem
Red team	Build a flawed artifact with an answer key, and find the defect in a classmate's	Find the defect planted in a classmate's project
Council of Agents	Student designs the roles	Roles supplied, run in class
Source audit	Five citations, one the student finds	Three supplied citations

Grading

AI is allowed, and expected, on all work outside the room. Half the grade comes from a midterm and a final, taken in class, on paper, without AI. The midterm covers Modules 1 to 3; the final covers all six. The other half comes from the term project and the in-class checks on it. An AI tool can produce take-home work, so most of the grade is earned in the room. Both versions use the same weights.

Component	Weight	Take-home, AI allowed	In class, AI-free, on paper
Examinations	50%	–	50% (midterm 20%, final 30%)
Module work	15%	5% completion	10% module checks
Term project and its rules	35%	20% graded submission	15% defense
Total	**100%**	**25%**	**75%**

Each module check gives students a copy of their module work with planted defects. They name each defect, the check that finds it, and a verdict.

Client data

Each client signs a one-page scope letter that says what the student may see and share. Projects use public, synthetic or client-approved data, and none collects data about people. A client is named only with written consent, never on a public page. Confidential work is graded on the check log in place of full session records.